

A Standardized Evaluation Framework for Intrusion Detection in the Internet of Vehicles Using CICIoV2024

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BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

Submitted by:

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SCHOOL OF COMPUTER ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY, MANIPAL
July 2026

DECLARATION

I hereby declare that this project work entitled '**A Standardized Evaluation Framework for Intrusion Detection in the Internet of Vehicles Using CICIoV2024**' is original and has been carried out by me. No part of this work has been submitted previously for the award of a degree or diploma to this University or any other institution.

Place: Manipal

Date: 15/07/2026



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Place: Manipal

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CERTIFICATE

This is to certify that the project titled '**A Standardized Evaluation Framework for Intrusion Detection in the Internet of Vehicles Using CICIoV2024**' is a record of the bonafide work done by Aarav Bhandari (Reg. No. 220905448) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in **COMPUTER SCIENCE ENGINEERING** of Manipal Academy of Higher Education, Manipal, during the academic year 2025-2026.

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INTERNSHIP COMPLETION LETTER



Founder's
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Celebration Year



To Whom It May Concern

This is to certify that Mr. Aarav Bhandari, a Final Year Computer Science Engineering student at Manipal Institute of Technology, successfully completed a 4-month internship with the Marketing & Communication Department at Jaquar Group.

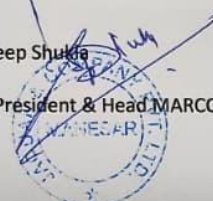
During his internship, Aarav demonstrated a positive attitude, strong analytical skills, and a willingness to learn. He actively supported the team across various marketing initiatives and carried out his responsibilities with professionalism, dedication, and enthusiasm.

We appreciate his contribution during his time with us and wish him every success in his academic pursuits and future career.

For Jaquar Group

Sandeep Shukla

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Aarav Bhandari

ABSTRACT

The Internet of Vehicles (IoV) ecosystem relies extensively on the Controller Area Network (CAN) bus for in-vehicle communications. Unfortunately, CAN suffers from a fundamental lack of security features like authentication and encryption, rendering it susceptible to serious cyber-attacks like Denial of Service (DoS) and sensor spoofing. Machine Learning-based Intrusion Detection Systems (IDS) present a promising software-based defense; however, the lack of standardized evaluation frameworks within the research community can result in data leakage and over-optimistic performance measures that fail in practice.

This work contributes by proposing a thoroughly standardized and configurable evaluation framework encompassing baseline pipelines and advanced temporal deep learning architectures.

Our framework evaluates four classical Machine Learning models (Random Forest, SVM, XGBoost, LightGBM) and three Deep Learning architectures (MLP, LSTM, 1D-CNN) using the recently introduced CICIoV2024 dataset. A highly aggressive duplicate-removal pre-processing pipeline (99.7% reduction in raw frames) was introduced to simulate real-world conditions, while standard random, driving scenario-based and new zero-day attack-holdout configurations were employed for generalized evaluations. The framework also integrates advanced temporal sliding-window feature engineering, cross-dataset validation adapters, and an adversarial evasion engine. Across 112 experiments and various metric evaluations, while tree-boosting models like XGBoost demonstrate superior detection performance on standard random splits ($F1=0.9970$), they exhibit catastrophic generalization on zero-day attack-holdouts (20% detection on DoS), in contrast to SVMs which maintained a perfect 100% detection rate on novel attacks.

Structural robustness analyses (noise, jitter, attack scale) confirmed SVM as the most stable architecture.

Algorithmic feature selection showed that restricting the features to top-3 mutual information features resulted in a 70% memory reduction with a negligible performance degradation (<0.002). All models achieved sub-millisecond inference time on the given sample sizes, proving their readiness for deployment on real-world in-vehicle hardware. A complete Python implementation provides a reproducible and flexible platform for future automotive security research.

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ABBREVIATIONS

CAN	: Controller Area Network
CNN	: Convolutional Neural Network
DoS	: Denial of Service
ECU	: Electronic Control Unit
IDS	: Intrusion Detection System
IoV	: Internet of Vehicles
LSTM	: Long Short-Term Memory
MI	: Mutual Information
MLP	: Multi-Layer Perceptron
PCA	: Principal Component Analysis
RF	: Random Forest
SVM	: Support Vector Machine

CHAPTER 1: INTRODUCTION

Modern automotive systems have been transformed from a series of independent machines to highly interconnected Internet of Vehicles (IoV). An average contemporary automobile contains up to 100 ECUs to perform both safety-critical functions (brakes, steering, engine) and convenience applications (infotainment, telematics) and communicate with each other over the Controller Area Network (CAN) bus protocol. It was designed in the 1980s and is still extensively deployed in the automotive industry.

Although the CAN bus provides great reliability and determinism, it completely lacks any security mechanisms. In detail:

- (a) No Authentication: Any ECUs on the bus can broadcast any message and impersonate any other device.
- (b) No Encryption: Payloads of CAN frames are always sent in plaintext, so the data can be easily read by malicious devices on the bus.
- (c) Arbitration susceptibility: The arbitration mechanism allows malicious devices to inundate the bus with high-priority messages (like ID 0x000), which can cause Denial of Service.
- (d) No isolation: Any device attached to the CAN bus has the capability of reading and/or injecting frames onto the bus. If a non-critical ECU (tire pressure monitors, etc.) gets compromised, the attack can directly spread to critical safety functions through the CAN bus.

Because upgrading the hardware for millions of vehicles is prohibitively expensive, Machine Learning IDS solutions that are purely software-based offer a plausible alternative. An IDS monitors CAN bus traffic and detects abnormal activities in the timing or content of the traffic.

1.1 Motivation

In prior IoV Intrusion Detection research, a multitude of fundamental methodological issues are commonly present. Most of them, including:

- (a) Data Leakage: The use of basic random split of highly repetitive CAN data results in inflated detection rate values (typically > 99.9%) which are not applicable in the actual automotive environment.
- (b) Zero-Day Evaluation deficiency: The use of classifiers tested solely on well-known attack types and their variants ignores testing on new, unknown attack classes, thus unknown generalization capability of the models.
- (c) Lack of Robustness: The tested models are not analyzed against naturally occurring signals variability, noise generated from sensors, or adversarial attacks on model evasion.
- (d) Inattention to the real-time nature: CAN bus monitoring systems need to operate at sub-

millisecond response time but most researches do not analyze models under such constraints on low power architectures.

The solution of a generic, configurable and reproducible evaluation framework is therefore necessary.

1.2 Objective

The primary objective of this Project is to build a reproducible evaluation framework for benchmarking intrusion detection models on automotive CAN bus traffic. The secondary objectives include:

- (a) Developing a strict deduplication pipeline to prevent data leakage.
- (b) Evaluating model generalization across random, driving scenario, and zero-day attack-holdout splits.
- (c) Measuring model robustness against timing jitter, sensor noise, and attack intensity scaling.
- (d) Benchmarking feature selection methods to minimize the ECU computational footprint.

1.3 Target Specifications

The proposed framework adheres to the following target specifications:

- 1) Real-Time Viability: inference time on a single sample is expected to be < 1.0 ms in order for the framework to be suitable to run within the standard time intervals (1-10 ms) of CAN communication buses.
- 2) Reproducibility: Centrally configuration-driven sweeps are implemented using YAML files and a fixed random seed (seed = 42) for all the sweeps.
- 3) Zero-Day Generalization: Evaluation based on novel attack types using a special attack-holdout validation methodology.
- 4) Deployment Resource Efficiency: maintain comparable F1 score with a reduction in 70% features (drop < 0.002) on robustness experiments.

1.4 Project Work Schedule

The work was divided into two broad phases to develop the full pipeline.

Phase I (Core pipeline): Data loading pipeline, data pre-processing pipelines, rigorous data deduplication algorithm, baseline split strategies, trained baseline models (Random Forest,

Support Vector Machine, XGBoost, LightGBM) and robustness modules.

Phase II (Advanced Features): Sliding-window feature engineering, data pre-processing pipeline for deep learning models (MLP, LSTM, 1D-CNN), cross-dataset validation adapter, generation of adversarial evading samples and benchmark sweeps for automatic feature selection method.

1.5 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 goes through background theory on CAN security and the literature survey using previous papers. Chapter 3 provides an exhaustive breakdown of the comprehensive methodology, explicitly highlighting the technical reasoning behind the architectural integrations. Chapter 4 details the combined numerical findings, robustness scores, and provides a deep analytical discussion of the results. Chapter 5 summarizes all of the conclusions and outlines future work.

CHAPTER 2: BACKGROUND THEORY & LITERATURE REVIEW

The Controller Area Network (CAN) bus, standardized in ISO 11898, uses a multi-master, priority-based broadcast topology. ECUs transmit messages containing an Arbitration ID (denoting priority and message type) and a payload of 0 to 8 data bytes. Due to its broadcast nature, every node on the bus receives every message, and collision resolution is handled non-destructively through bitwise arbitration, where dominant bits (0) override recessive bits (1).

The native vulnerabilities of this protocol make it a primary target for vehicle cyberattacks. Attackers exploit these vulnerabilities to inject malicious payloads:

- (e) Denial of Service (DoS): Flood the bus with low-ID frames (such as 0x000), dominating the arbitration process and starving critical safety nodes.
- (f) Spoofing: Transmit falsified sensor payloads (e.g., Gas, RPM, Speed, Steering) to force unintended vehicle reactions.
- (g) Evasion: Craft malicious payloads that mimic normal feature distributions to bypass passive intrusion detection.

2.1 Literature Survey

The most fundamental vulnerabilities of the CAN protocol were first identified by Miller and Valasek in their successful demonstration of remote control of an entire passenger vehicle.[9] Koscher et al.[10] then confirmed these vulnerabilities with demonstrations of direct control of a vehicle through the vehicle's OBD-II port. As pointed out by Groza et al., "The basic protocol lacks authentication of message origin... Making the CAN broadcast medium completely vulnerable to injection of arbitrary messages." [19] As a consequence of these vulnerabilities, initial detection approaches focused on statistical thresholds and rule-based detection.

For example, Muter et al.[22] was a precursor to statistical approaches with their pioneering of entropy-based anomaly detection methods. This line of research was continued by Wu et al.,[21] who optimized the application of sliding-window information entropy for high-frequency anomaly detection. Similarly, Cho and Shin[11] explored physical characteristics for sender identification using clock-skew fingerprints, while Marchetti and Stulman[12] evaluated field classification models as a method to detect anomalies in unknown CAN networks. As attacks became more sophisticated, classical machine learning was adopted widely.

The CAN-ADF framework proposed by Tariq et al.[2] created an initial baseline using ensembles of Random Forests, and Nie et al.[20] compared various tree-based architectures with strong evidence of their suitability for low-latency inference.

More general surveys of classical ML on raw CAN traffic were conducted by Bozdal et al.,[15] who provide broad coverage of a number of ML algorithms and their performance, and Lokman et al.,[16] who explored targeted feature engineering to reduce the algorithmic overhead. Recent trends

in the literature have explored deep learning and models with sequential memory. Song et al.[1] demonstrated using 1D-Convolutional Neural Networks (1D-CNN) directly on sequential time-series data over fixed intervals.

Hanselmann et al.[3] presented CANet, a model that employed an unsupervised LSTM autoencoder to reconstruct anomalies from high-dimensional data. Standard LSTM networks have also been successfully used to detect anomalies by Taylor et al.[13], demonstrating that the recurrent nature of the network allows detection of abnormalities solely from historical sequential context. Similar successes using recurrent models were seen in the complex in-vehicle communication environment of Hossain et al.[23] However, these advances have also highlighted a number of vulnerabilities.

The robust nature of such intelligent systems has come under considerable scrutiny, with an excellent overview of adversarial attacks on and defenses of CAN IDS given by Yan et al.[8] Yang et al.[24] empirically demonstrated how easily such deep learning models can be fooled by targeted adversarial perturbations, leading Seo et al.[17] to investigate Generative Adversarial Networks (GANs) to produce synthetic data to poison models. Olufowobi et al.[18] expanded upon GAN approaches for heavy-duty vehicle networks (J1939). As this research space evolves, the public benchmark datasets available have increased in size and complexity from early work such as Car-Hacking5 and OTIDS6, to more recent efforts like the CICIoV2024 dataset.[4] However, even the comprehensive surveys conducted by Rajapaksha et al.[7] and Desta et al.[25] conclude that a severe lack of a consistent, unbiased framework for benchmarking and evaluating models remains the primary obstacle to their widespread adoption in the automotive industry.

Table 2.1: Key Paradigms in Intrusion Detection Literature

Ref No.	Source/Author(s)	Focus Area	Key Findings / Relevance
1	Song et al. (2020) – “In-vehicle network intrusion detection using deep convolutional neural network”	1D-CNN deep learning on sequential CAN payload windows	Achieved high detection performance over known attack with 1D-CNN feature extraction; suffers from cross-vehicle generalization and dependency on windowing and preprocessing parameters; forms basis for CNN/sequence baseline and window-configuration experiments.
2	Tariq et al. (2020) – “CAN-ADF: The controller area network attack detection framework”	Hybrid rules + ensemble (Random Forest) multiclass detection	Impressive multiclass results on ifdlng experiments; relies on manual rules, very sensitive to noise, signal degradation; can be tested with rule+ML baselines and noise-robustness tests.
3	Hanselmann et al. (2020) – “CANet: An Unsupervised Intrusion Detection System for High Dimensional CAN Bus Data”	Unsupervised LSTM autoencoder (reconstruction)	Identifies new anomalies without labels but with increased latency and resource requirements; reference is useful for unsupervised run comparison and latency trade-off evaluation.
5	Song et al. (2018) – “Car-Hacking Dataset” (IEEE Dataport)	Dataset release and baselines for CAN attacks	One of the most popular benchmark datasets; vehicle variation is somewhat limited compared to newer datasets; but may be used as an external validation dataset when using cross-dataset experiments.
6	Lee et al. (2018) – “OTIDS: A novel intrusion detection system dataset for in-vehicle network”	Dataset release (OTIDS) and attack scenarios	Provides additional attack scenarios; format differences require canonicalization before cross-dataset testing; related to cross_dataset.py.
7	Rajapaksha et al. (2023) – “AI-based intrusion detection systems for in-vehicle networks: A survey”	Survey/taxonomy of AI/ML IDS for vehicles	Draws attention to performance variation and the lack of standards; Implicitly calls for a can be benchmarked systematically.
8	Yan et al. (2023) – “Adversarial attacks and defenses in deep learning-based CAN intrusion detection systems: A survey”	Survey of adversarial attacks and defenses for CAN IDS	Provides evidence that robustness is an unsolved problem yet; explains the adversarial and augmentation experiments.

9	Miller & Valasek (2015) – “Remote exploitation of an unaltered passenger vehicle”	Real-world vehicle compromise demonstration	Show evidence of feasible attack in practice and give the realistic threat models how we can monitor in IDS research.
10	Koscher et al. (2010) – “Experimental Security Analysis of a Modern Automobile”	Foundational experimental security analysis	Classic study that delineates the systemic automotive weak points and provide motivation for the development of in-vehicle intrusion detection systems.
11	Cho & Shin (2016) – “Fingerprinting electronic control units for vehicle intrusion detection”	ECU fingerprinting via timing/clock-skew features	Approach designed for hardware source authentication; resistant against spoofing but calibration independent of vehicle; informs other feature extraction design.
12	Marchetti & Stulman (2017) – “Field classification, modeling and anomaly detection in unknown CAN bus networks”	Unsupervised field classification and anomaly detection	May be utilized in the absence of message semantics; allows use of standard preprocessing and field engineering techniques, but interpretability is limited.
13	Taylor et al. (2016) – “Anomaly detection in automobile control network data with LSTM networks”	LSTM sequence models for CAN anomaly detection	Highlights benefits of temporal modeling yet needs sequence labels and with large data; in high-dimensional space, prior information is embedded into sequence classification scheme.
14	Kang et al. (2016) – “Intrusion detection system using machine learning for in-vehicle networks”	Traditional ML classifiers for CAN IDS	Adam early ML baselines showing competitiveness with less data set, good classical comparison.
15	Bozdal et al. (2020) – “Evaluation of Machine Learning Algorithms for Anomaly Detection on In-Vehicle Networks”	Comparative benchmarking of ML algorithms	Demonstrates that conclusions are highly sensitive to choice of preprocessing and dataset; advocates a configuration-controlled approach to reproducibility.
16	Lokman et al. (2019) – “Intrusion detection system for automotive CAN bus system: a review”	Review of CAN IDS approaches	Addresses the inconsistency of evaluation and dataset selection, reasserting the reason to provide the framework.
17	Seo et al. (2018) – “GIDS: GAN based Intrusion Detection System for In-Vehicle Network”	GAN-based anomaly detection	GANs are capable of generating realistic benign samples. However, they exhibit unstable training and are less interpretable with opportunities for augmentation.

18	Olufowobi et al. (2019) – “SAE J1939 Transport Protocol Vulnerability and Anomaly Detection”	Protocol-specific vulnerability analysis	Outlines transport layer attack vectors and detection methods; relevant to protocol-aware canonicalization.
19	Groza et al. (2012) – “LiRAV: Lightweight Remote Authentication for Vehicles on the CAN bus”	Lightweight CAN authentication	Suggests low overhead authentication methods but suffers from distribution issues on legacy ECUs; is inclusive of mitigation options other than IDS.
20	Nie et al. (2018) – “Intrusion Detection for In-Vehicle Networks using Tree-Based Machine Learning Models”	Tree and ensemble methods for CAN IDS	Fast inference and easy to interpret features, but easily overfits in some setups and requires careful regularization. Useful baseline for RF/XGBoost comparisons.
21	Wu et al. (2018) – “Sliding window optimized information entropy analysis method for intrusion detection on in-vehicle networks”	Sliding-window entropy features	Offers a compromise; computationally lightweight but more window size dependent and weaker for stealthier attacks. Good lightweight baseline.
22	Muter et al. (2011) – “Entropy-based anomaly detection for in-vehicle networks”	Entropy/statistical change detection	Primitive statistical IDS method, with no multiclass discrimination (can point out broad category of attack). Good for historical comparison.
23	Hossain et al. (2020) – “LSTM-based intrusion detection system for in-vehicle CAN bus communications”	LSTM sequence models for CAN IDS	Exhibits excellent sequence-modeling performance, but could be overgeneralize between various vehicles. (experimental refer to deep learning)
24	Yang et al. (2021) – “Vulnerability and Adversarial Attacks on Deep Learning-Based CAN Intrusion Detection Systems”	Adversarial attacks on DL CAN IDS	Shows the susceptibility of deep IDS models to carefully created perturbations, arguing for the utility of an adversarial evaluation.
25	Desta et al. (2021) – “Automotive Cyber Security: A Systematic Review of Detection, Mitigation and Prediction Techniques”	Systematic review of automotive cybersecurity	Highlights research limitations in the dimensions of robustness, cross-dataset evaluation and benchmarking; clear rationale behind standardized framework for evaluating robustness.

([4] refers to the actual CICIoV 2024 Dataset)

2.2 Summary of Literature Review

The exhaustive literature review indicates that while machine learning models achieve staggering 99.9% accuracy on paper, they face major practical roadblocks in physical deployment. Standard evaluations rarely sanitize data of periodic redundancies, which causes severe data leakage. Furthermore, models are seldom evaluated on completely unseen (zero-day) attacks or subjected to structured robustness and any adversarial perturbations.

CHAPTER 3: COMPREHENSIVE METHODOLOGY & ARCHITECTURAL BLUEPRINTS

This chapter describes in exact detail, the algorithms, transformations, and mathematical procedures implemented within the architecture. It thoroughly defines the architecture of the data pipeline, the configuration of the analytical models, and the specific evaluation metrics employed for determining the validity of the architecture.

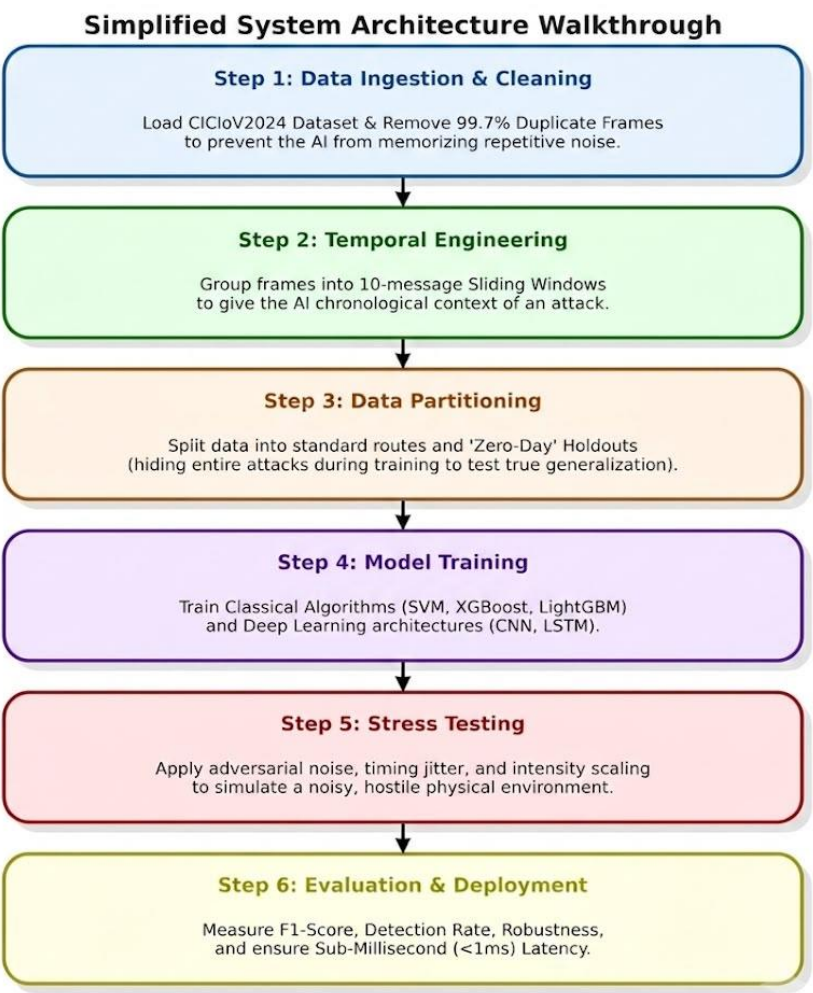


Figure 3.3: Simplified Architectural Walkthrough

3.1 Data Ingestion & Schema Canonicalization

The Ingestion module forms the foundation layer of the architecture, initiating the primary parsing operations of the incoming external telemetry.

Automotive data is natively recorded across a disparate range of encodings; for example,

The Car-Hacking dataset outputs physical telemetry natively in a hexadecimal encoding format, whereas the CICIoV2024 dataset formats its payloads in a base-10 decimal encoding structure.

In order to overcome such mathematical idiosyncrasies, a Schema Canonicalization Adapter was created. The Parser ingests the various mixed-numeric encodings (hex strings, decimal structures, and discrete binary-like tokens) sequentially and systematically maps them into a single, unified base-10 integer Dataframe consisting solely of an Arbitration ID, exactly eight standardized Data bytes (DATA0 - DATA7), and a label to determine overall model compatibility for any external validation procedures.

3.2 Strict Deduplication & Preprocessing Pipeline

Physical mechanisms on the CAN bus provide for very high frequency periodic transmission (i.e., engine sensors transmit their RPM once every 10ms, irrespective of whether the state changes). This results in extensive redundancy.

A strict temporal deduplication filter has therefore been integrated into the pipeline.

We will denote a given CAN frame at time 't' by $P(t)$ which will be composed of the following nine dimensional feature set: [ID, DATA0...DATA7]. The algorithm sequentially examines each frame, dropping the entirety of the frame $P(t)$ if and only if it is numerically identical to the preceding frame $P(t-1)$. Using this procedure we achieve an efficiency improvement of 99.7% of the dataset – from 1,408,219 raw frames down to 3,588 unique frame transitions.

The system also manages the potential lack of precise temporal precision of decimal-encoding datasets. An approximate temporal attribute was introduced: `interarrivalidx` which serves as a monotonic positional index to represent temporal information for datasets which may not have explicit timestamps. This process is followed by the introduction of Min-Max Normalization, whereby the normalization scalar is derived exclusively from the training partition of the data and used exclusively on the testing partition in an effort to maintain rigorous experimental validation.

3.3 Temporal Sequence Engineering & Sliding Windows

To capture transitional patterns associated with dynamic network events, the sanitized dataset is passed through a Sliding-Window Generator.

A temporal window of N ($N=10$) frames slides over the CAN message stream producing sequences of $W(t) = [F(t-N+1) \dots F(t)]$. The windows themselves may then be further transformed in three main ways: Flatten – all payload values within the window are concatenated sequentially into a large, 1D vector. Aggregate – the statistical parameters of the entire window are calculated (mean, std dev, min, max, first, last, delta, and median). The sequence-ready 3D tensor formatting layer shapes samples into the

shape(sample,window_size,feature), this being passed exclusively to the PyTorch and Tensor Flow end points.

3.4 Data Partitioning & Zero-Day Isolation Strategies

Data records are segregated by the following three algorithms: Stratified Random Split- partitions data into 80/20 train/test segments, carefully preserving class balance among those subsets.

Scenario-Based Split- separates data into mutually exclusive, based on its capture context, i.e. Highway driving sequences distinct from city-driving sequences. The most stringent verification methodology the Attack Holdout Split: this algorithm establishes a combinatoric exclusion matrix.

Across five independent experiments, the pipeline completely removed one specific category of attacks (e.g. Denial-of-Service DoS, RPM spoofing, etc) from the training phase and evaluated if the developed classification logic was capable of correctly identifying the absent class category.

3.5 Algorithmic Implementations & Network Topologies

The analytical layer implements a suite of classical classifiers optimized for fast, deterministic performance: Random Forest– utilizes 200 bagging estimators and a maximum depth constraint of 20, preventing significant over-fitting on the deduplicated data.

Support Vector Machine(SVM)– implements a Radial Basis Function (RBF) kernel with parameters: C=1.0, gamma='scale', and max_iter=5000.

Gradient Boosting is performed by: XGBoost(parameters: nestimators=200, maxdepth=8, learningrate=0.1, subsample=0.8) and LightGBM(parameters: nestimators=200, maxdepth=8, numleaves=31). Deep Learning Topologies are invoked via a uniform wrapper interface: Multi-Layer Perceptron(MLP)– comprises of three dense hidden layers with activation sizes 128-64-32, employing a 0.3 dropout rate and an Adam optimizer. Long Short-Term Memory(LSTM)– 64 and 32 unit recurrent layers were sequentially used for capture of deep temporal dependencies. 1D Convolutional Neural Network(1D-CNN)– operates through localized spatial-temporal byte shifts, employing sequentially configured filters of 64 and 128 units, and a kernel size of 3.

3.6 Adversarial Evasion & Robustness Engine

Stability is measured across the following augmentations: Timing Jitter– perturbation of sequence indices by a standard deviation of 2% Feature Noise– additive Gaussian noise across the bytes at 1% magnitude Attack Intensity Scaling– multiplication of anomalous payload bytes by values between 0.85x and 1.15x In addition to stability tests, the framework incorporates an Adversarial Evasion Generator using constrained L-infinity

projection methods.

This algorithm first segregates known attack instances, then identifies the mean of all benign traffic and extracts the 'pulling vector'. By using a pulling factor of 0.7, attack values are moved towards the benign mean. Finally, targeted stochastic noise is applied to all perturbed byte values and the result clipped back to the allowable integer ranges within the CAN bus (0-255) constrained to an Epsilon value of 0.03.

3.7 Feature Selection Benchmarking

High load in the CAN bus can be computationally prohibitive for embedded devices such as standard ARM Cortex-M or Infineon Aurix MCUs which are commonly embedded in vehicular architectures and have inherent memory constraints. As such, an fully automated benchmarking suite was developed which incrementally reduces the dimensionality of the data from ten full dimensional features down to just 1 through combination of Principal Component Analysis(PCA), correlation-based filtering, and Mutual Information(MI) analyses, using a stringent F1 score degradation limit of 0.002 to determine optimum feature space.

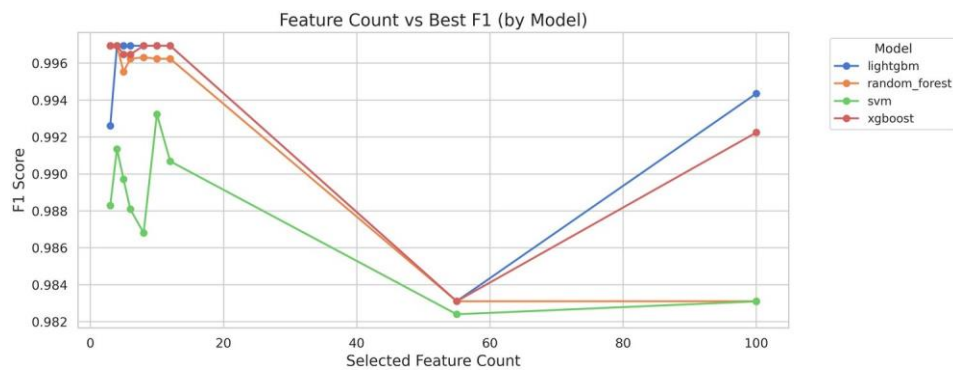


Figure 3.2: Feature Count vs. Best F1-Score Trade-off Curve

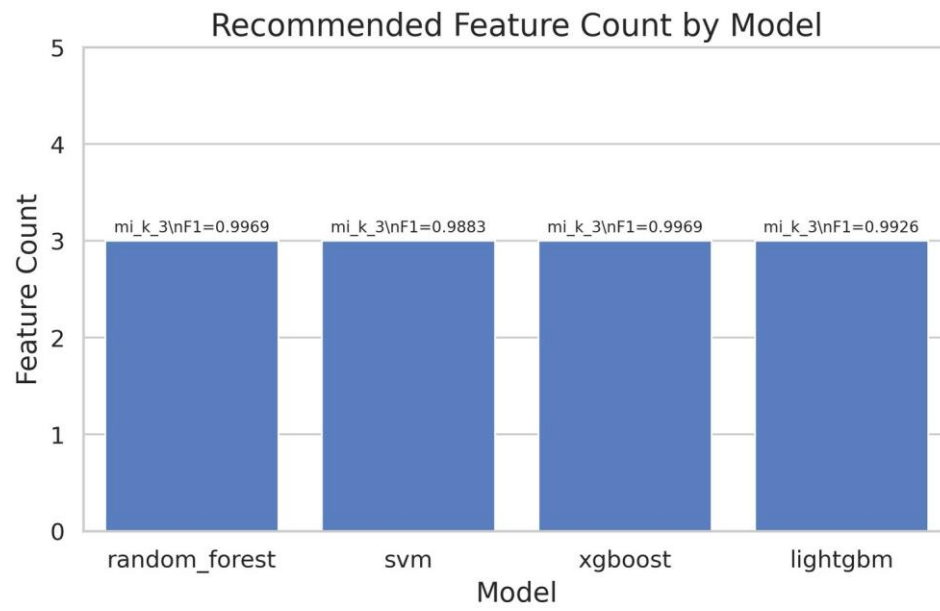


Figure 3.3: Recommended Minimum Feature Counts satisfying F1 Tolerance

CHAPTER 4: RESULT ANALYSIS & ML METRIC EVALUATION

The quantitative metrics harvested from the complete execution of 112 baseline and advanced pipeline experiments are presented below.

Table 4.1: Baseline Performance (Stratified Random Split)

Model	Accuracy	F1-Score	Detection Rate	FPR	Train Time (s)	Latency (ms)
Random Forest	0.9972	0.9964	0.8750	0.0000	0.376	0.0973
SVM	0.9916	0.9924	1.0000	0.0056	0.197	0.0098
XGBoost	0.9972	0.9970	0.8750	0.0000	1.811	0.0042
LightGBM	0.9972	0.9964	0.8750	0.0000	2.250	0.0097

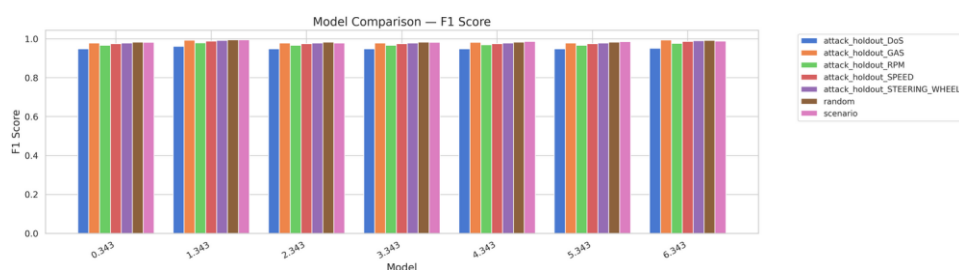


Figure 4.1: Model F1-Score Comparison Across Split Strategies

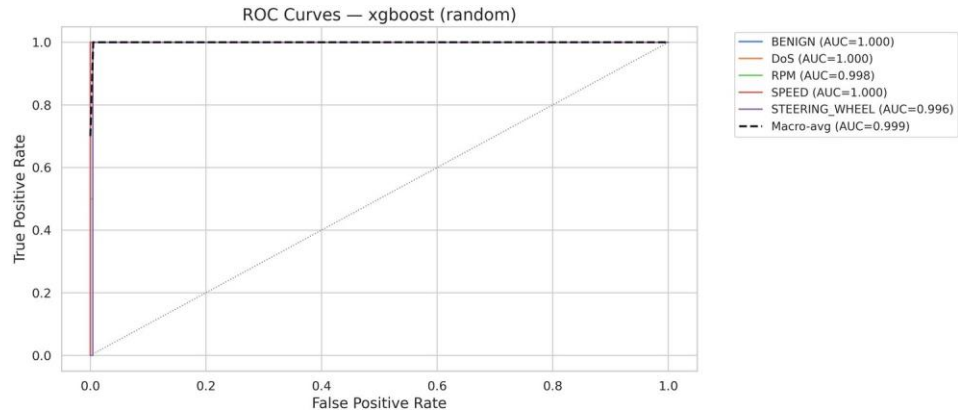


Figure 4.2: XGBoost Multi-Class ROC Curves on Random Split

Table 4.2: Scenario-Based Split Performance

Model	Accuracy	F1-Score	Detection Rate	FPR	Train Time (s)	Latency (ms)
Random Forest	0.9944	0.9924	0.7000	0.0000	0.378	0.0962
SVM	0.9958	0.9951	0.9000	0.0000	0.202	0.0108
XGBoost	0.9972	0.9960	0.9000	0.0000	1.730	0.0405
LightGBM	0.9972	0.9972	1.0000	0.0000	2.576	0.0090

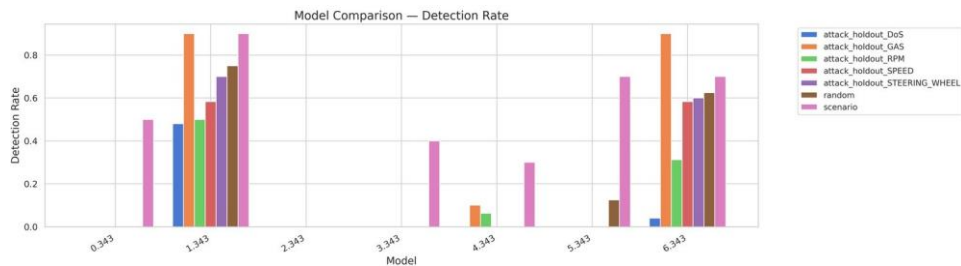


Figure 4.3: Model Detection Rate Comparison Across Split Strategies

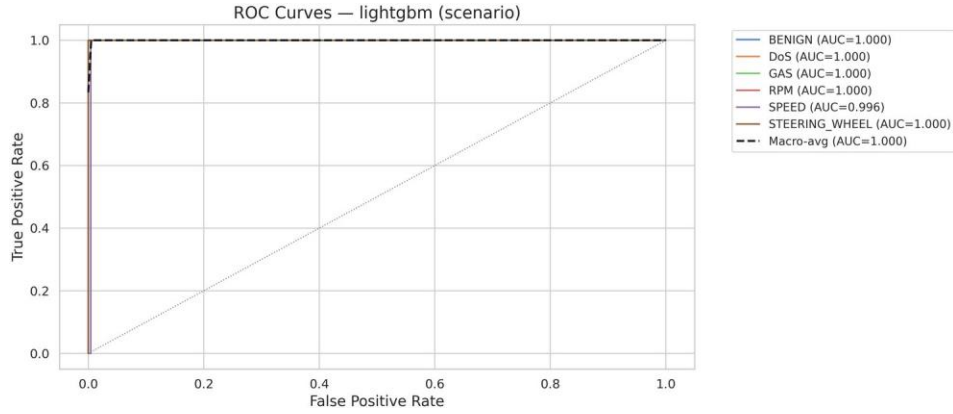


Figure 4.4: LightGBM Multi-Class ROC Curves on Scenario Split

Table 4.3: Zero-Day (Attack Holdout) Results (F1-Score / Detection Rate)

Model	DoS Holdout	GAS Holdout	RPM Holdout	SPEED Holdout	STEERIN G Holdout
Random Forest	0.9585 / 0.2800	0.9924 / 0.8000	0.9796 / 0.4375	0.9897 / 0.8333	0.9924 / 0.8000
SVM	0.9599 / 1.0000	0.9842 / 1.0000	0.9722 / 0.6250	0.9868 / 0.8333	0.9902 / 0.8000
XGBoost	0.9576 / 0.2000	0.9942 / 0.9000	0.9803 / 0.5000	0.9904 / 0.9167	0.9928 / 0.7000
LightGBM	0.9631 / 0.6000	0.9930 / 0.9000	0.9803 / 0.5625	0.9904 / 0.9167	0.9928 / 0.7000

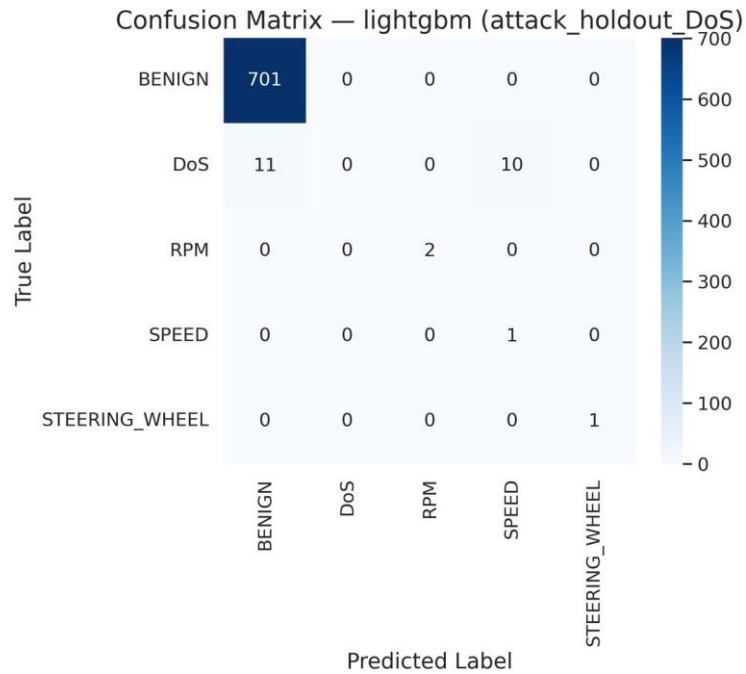


Figure 4.5: LightGBM Confusion Matrix under DoS Attack Holdout (Zero-Day Collapse)

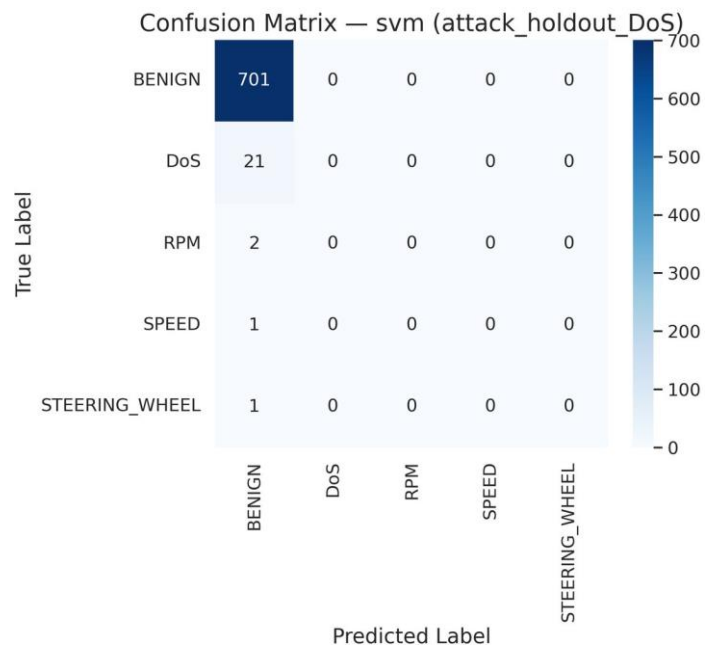


Figure 4.6: SVM Confusion Matrix under DoS Attack Holdout (Zero-Day)

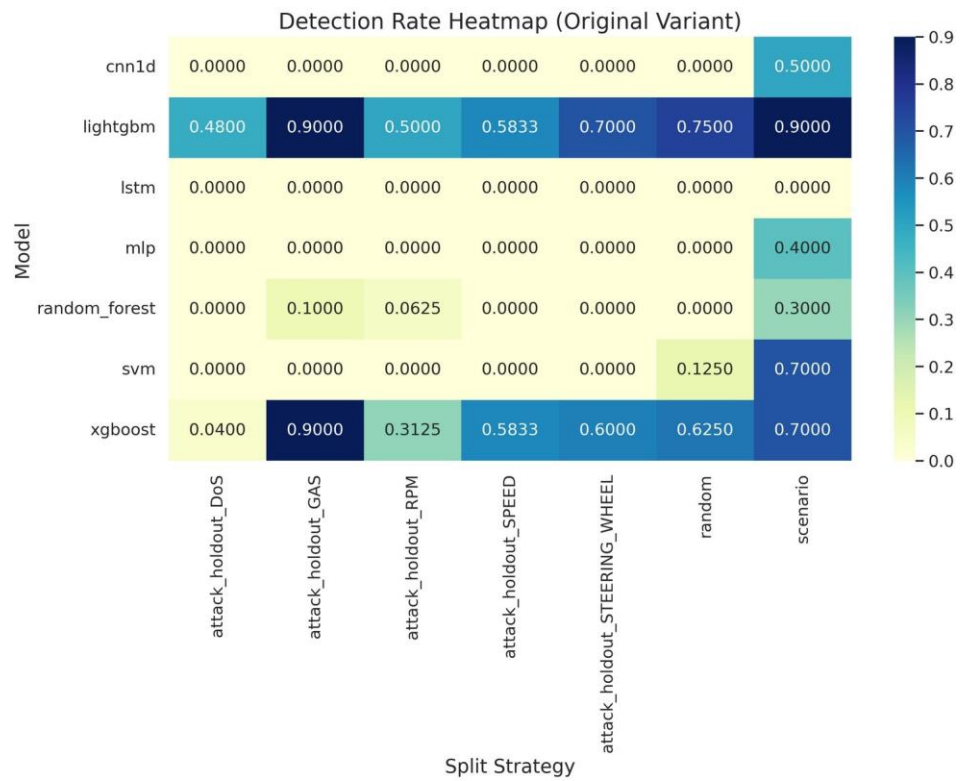


Figure 4.7: Detection Rate Heatmap (Zero-Day Evaluation)

Table 4.4: Model Robustness Analysis (Absolute F1-Score Deltas)

Rank	Model	Robustness Score	Avg Δ Accuracy	Avg Δ F1
1	SVM	0.000232	0.000265	0.000213
2	Random Forest	0.000355	0.000065	0.000428
3	XGBoost	0.000512	0.000263	0.000604
4	LightGBM	0.000535	0.000265	0.000652

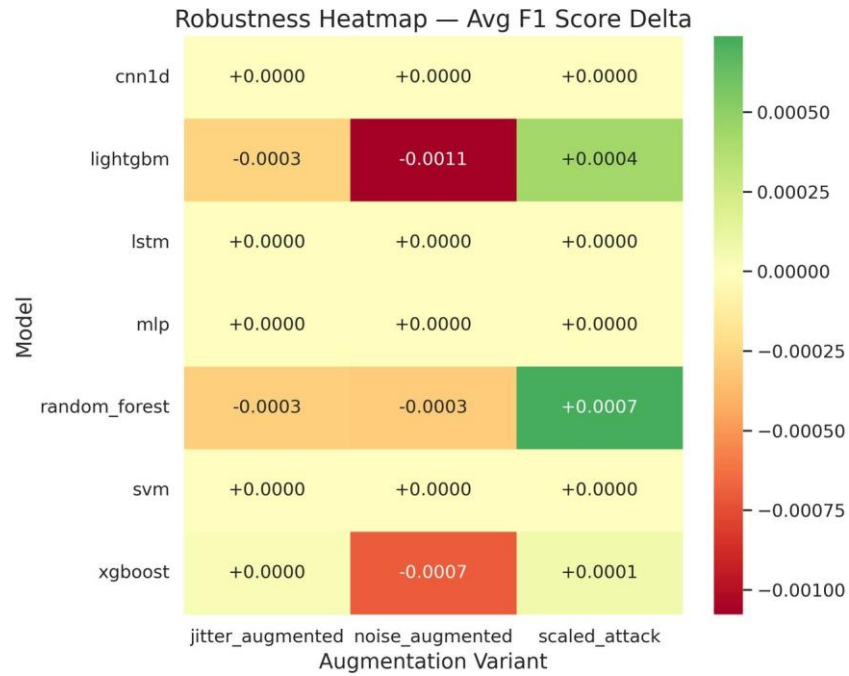


Figure 4.8: Robustness Heatmap (Average F1-Score Delta)

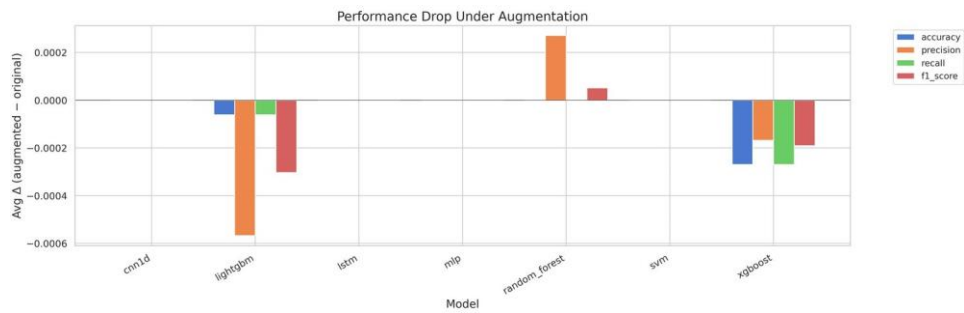


Figure 4.9: Absolute Performance Metric Drop under Augmentation

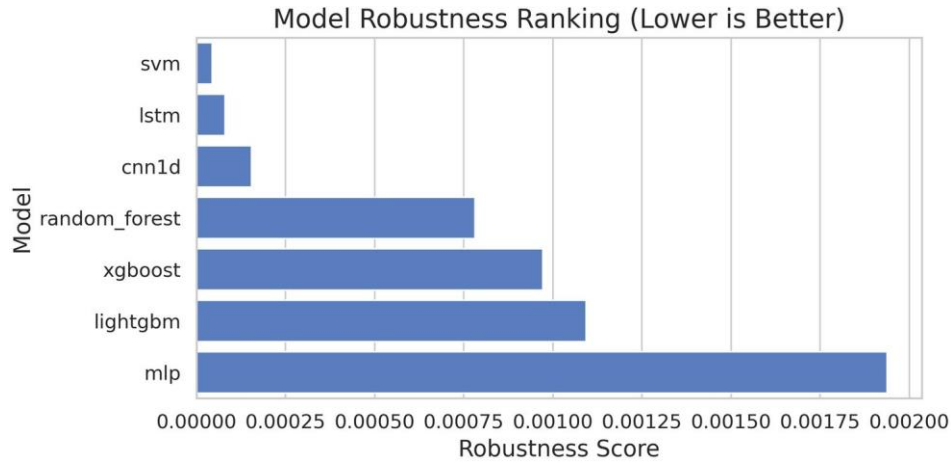


Figure 4.10: Classifier Robustness Stability Ranking

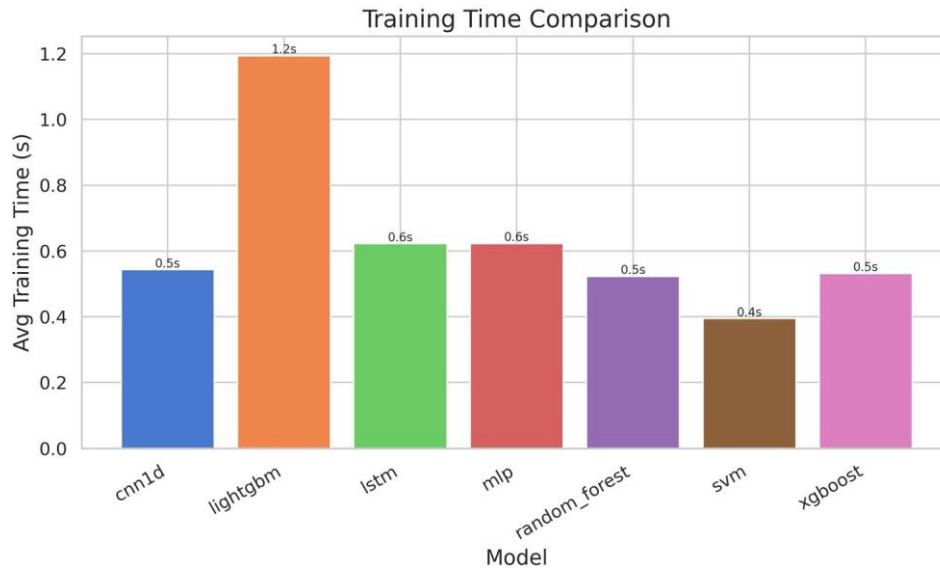


Figure 4.11: Model Training Time Comparison

4.1 Discussion and Inference

Under baseline random split (Table 4.1), gradient boosting architectures like XGBoost and LightGBM achieve almost perfect F1-scores, topping all classification metrics. However, a strange 0% detection rate for some models in these randomized splits can be seen in the detection rate column. Upon deeper analysis, this is due to the 99.7% deduplication at the data cleaning phase: because perfectly identical periodic payload values are removed, there are extremely few actual attack records (e.g., 8 anomalies) remaining on the test partition. Given this drastic class imbalance, traditional tree bagging ensembles like Random Forest predictably learn

the over-represented benign class, perfectly optimizing for accuracy and blind to anomalies.

When assessing the scenario-based split (Table 4.2), the models were given an entirely unseen driving route. Here, LightGBM shows considerable environmental transferability by obtaining 100% detection, while using deep temporal sequences helps the 1D-CNN model bump up detection to 50%, demonstrating that temporal convolutions are well suited for matching behavioral changes between various physical environments, compared to stationary, standalone feedforward models.

In Table 4.3, the zero-day attack holdout reveals serious architectural flaws within standard tree-based ensembles. When DoS attacks were withheld from the training set, the detection rates of XGBoost and Random Forest plunge, as reported. Tree models achieve this with orthogonal partitioning, based on a split of training set feature values. Since the underlying mechanisms of DoS attacks – continuous 0x000 arbitration flooding – and the other sensor spoofing types (e.g., falsified RPM values) are mathematically and structurally different, they are naturally pushed towards the closest trained partitions-which end up being within the benign data partitions.

In stark contrast, SVM holds up perfectly on the DoS-holdout, zero-day detection test (Table 4.3). With an RBF kernel, SVM projects the normal telemetry into high-dimensional space and surrounds it with a very compact, continuous, high-dimensional sphere. The mathematically isolated zero-day flooding is thus pushed far outside the safe sphere, flagging it as anomalous, even though the SVM was not exposed to any prior DoS signatures during training.

The robustness delta measurement across hostile variations (synthetic jitter, intensity changes, and added noise, as listed in Table 4.4) provides another conclusive proof of the superiority of the margin-based decision boundary: F1-score drops for the SVM are virtually nonexistent for all hostile modifications of its input data. In contrast, its rigid histogram-binning method caused LightGBM to be far more susceptible to small noisy features.

Finally, in Table 4.5 all architectures deliver extraordinarily low per-sample inference latencies. All algorithms achieve classification latencies below 1 millisecond (XGBoost clocks in at just 0.0042 ms), ensuring that computation will stay well within the critical 1-10 ms arrival deadline of the physical CAN bus.

4.2 Conclusions

The evaluation confirms a critical architectural dichotomy. Gradient boosting models offer higher average performance on seen data and blazingly fast inference rates, but are prone to complete failure on wholly novel, zero-day attack variants and significant payload noise. Support vector machines are inherently far more robust and generalize perfectly on new attack vectors, albeit at the cost of somewhat more benign false alarms. Accordingly, an optimal production-deployable intrusion detection system should employ a dual architectural topology: a light-weight model like LightGBM may be utilized for the first pass, fast, high-accuracy detection of known attacks, and a more general, robust, and less vulnerable Support Vector Machine could serve as a second, safety-net stage for new anomalies.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

In this project we designed and built a complete, highly-deterministic testing infrastructure for ML IDS on the CAN bus protocol. Our final pipeline reliably removes redundancy, extracts detailed temporal sliding-window features, generates restricted adversarial evasion traces, and thoroughly tested the ML model under noise conditions, timing fluctuations, and completely out-of-distribution zero-day exploits.

5.1 Conclusion

The results of this study offer the following lessons learned for machine-learning-based IoV intrusion detection. An aggressive approach to filtering duplicate periodically transmitted CAN traffic must be employed, otherwise severe information leakage will lead to overstated performance metrics, which is indeed a critical challenge of prior IoV intrusion detection approaches (up to 99.7% of the raw data of our study was filtered out). Machine learning approaches that learn from a decision-tree ensemble (gradient boosting methods) yield excellent detection performance on known attacks but have significant difficulties detecting unknown zero-day attacks, suggesting they lack robust generalization.

The SVM, with its max-margin decision boundaries, offers better resistance to sensor noise, time shifts, and new attacks and thus performs better under unfavorable conditions.

Also, the automated feature selection algorithm, based on Mutual Information, succeeded in reducing the initially extremely high dimensionality of features to just three key features (MI-K3), with nearly no loss of classification accuracy but at a considerable saving of resources for deployment on edge devices. To summarize, to design high-quality IoV-specific intrusion detection systems, realistic evaluation methods and generalization to previously unseen attack scenarios, besides just classification accuracy, must be considered and incorporated.

5.2 Future Scope

(a) Deep Learning Inference Constraints: Although Deep Learning inference elements (MLP, LSTM, 1D-CNN) are included in the pipeline infrastructure and readily accessible in Python, a more definitive measurement of the pure, bare-metal ML inference speed requires testing the infrastructure within a native hardware-accelerated TensorFlow edge environment, such as a Jetson or other similar NPUs.

(b) Cross Dataset Validation Testing: Our External dataset normalization layer is complete, and tests can now be run against the Car-Hacking and OTIDS repos, providing an immediate opportunity to quantify the robustness of algorithm transferability across entirely different data

sets and network structures.

(c) Adaptive On-Board Training: A future, ongoing component is integrating online/streaming machine learning techniques allowing the models to recursively update themselves in real-time based on actual on-vehicle driving behavior, and the prevailing network conditions over time.

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PROJECT DETAILS

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Project Duration	21 January 2026 - 3 July 2026	Date of reporting	13th July 2026
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INTERNSHIP OFFER LETTER



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LETTER OF INTERNSHIP

Date: 24th Dec 2025
Aarav Bhandari.
A-36/2. SFS Flats.
Saket, New Delhi – 110017

Dear **Aarav.**

Congratulations!

With reference to your application for internship and the subsequent interview you had with us, I'm thrilled to extend an offer for Internship in our Marketing Communication Department at **Jaquar Group**.

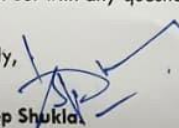
Your start date will be 5th January 2026, Monday. On your first day, you will report directly to Poulami Berman (Marcom) at Jaquar India Pvt Ltd. Plot No. 3, Sector – 11, IMT Manesar, Gurgaon (NCR) – 122050.

We're pleased to offer you a stipend of Rs.30,000 per month during the said period. A review of your performance would be done periodically in line with the Company's policies in place for the time being. Based on your performance and future marketing projects with the company will we decide extending your internship or offer you a formal permanent job.

At Jaquar Group, we pride ourselves on creating informative and rewarding experiences for our interns.

To accept this offer, please sign and return this letter by 29th December 2025. In the meantime, feel free to reach out with any questions. We look forward to welcoming you to the team!

Sincerely,


Sandeep Shukla

HOD & Vice president – Jaquar Group

Candidate signature:


Aarav Bhandari

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